COURSE GUIDE & SYLLABUS · GRADES 6–10 · AGES 11–15

Python (Gr 6–8) Beginner: Python Game Coders

Type real Python from day one and build a playable program in every class.

A first course in real, typed Python for ages 11–15. In eight live classes learners go from print("Hello") to a full text adventure game. Every class ends with a program they can play and show at home: a name badge or cricket scoreboard poster, a Mad Libs story generator, a UPI shop calculator, a Quiz Show, a rocket launch and times-table trainer, Guess the Number, Rock-Paper-Scissors and a Treasure Hunt adventure. Each lesson has a short video that shows the code being typed, a build-along, auto-checked Python Lab challenges, a predict-the-output puzzle, a debugging challenge, a worksheet and a full teacher lesson plan.

What learners will be able to do

- Write, run and fix Python programs using print, strings and comments
- Store information in variables and ask the user questions with input()
- Build sentences with f-strings
- Do maths with int() and the operators + - * / // %
- Make decisions with if, elif and else and comparison operators
- Keep score in a game with a counter variable
- Repeat code with for loops, range() and while loops, and fix infinite loops
- Use the random module to make games with surprises
- Store and use many values in a list with index, append, len and in
- Define and call a simple function with def
- Read error messages and debug programs step by step
- Plan, build, playtest and present an original text adventure game

How every lesson works

Watch	A short animated explainer video with captions and read-aloud narration.
Learn	Learn cards with real Python and a type-along build.
Python Lab	Write and run real Python in the browser. Challenges are checked automatically by running the program; hints unlock one at a time.
Quick checks	Predict-the-output and debugging questions with instant feedback.
Probe	Debugging and “new situation” tasks to show real understanding.
Projects	A playable game or tool every module. Starter code for learners and a model solution for the teacher.
Printables	Worksheet for every lesson, project brief for every project and a teacher lesson plan with answer keys.

Scope and sequence

#	Lesson	Big idea	Learners build
1	**Hello, Python!** Write your first Python programs with print(), strings and comments, and make text art that shows on screen.	print, strings, comments	Build along: my name badge
2	**Variables and Input** Store information in variables, ask the user questions with input(), and build sentences with f-strings.	variables, input, f-strings	Build along: friendly greeter
3	**Numbers and Decisions** Turn typed answers into numbers with int(), do maths in Python, and make decisions with if, elif and else.	int, maths, if/elif/else	Build along: cinema ticket counter
4	**Project: Quiz Show** Build a five-question quiz game that keeps score, gives feedback on every answer and awards a final grade.	score variables and decisions	Project: Quiz Show
5	**Loops: Repeat Like a Robot** Repeat code with for loops and range(), keep going with while loops, and avoid infinite loops.	for, range, while	Build along: loop practice
6	**Random and Games** Use the random module to roll dice and pick choices, and build a guessing game with a while loop, hints and an attempt counter.	random, game loops	Build along: dice roller
7	**Lists: Many Things in One Box** Make lists, read items by their index, add items with append, loop through a list and check what is in it.	lists, index, append, in	Build along: my cricket team list
8	**Capstone: Treasure Hunt** Design and build a text adventure game with rooms, choices, an inventory list, win and lose endings, and your first functions.	game design, lists, def	Project: Treasure Hunt Text Adventure

Modules

Module 1 · Talk to the computer

print, strings, variables, input, f-strings, numbers and decisions. Project: Quiz Show.

Big question: How do we give a computer instructions, and how can it answer back?

Module 2 · Loops and games

for and while loops, random numbers, lists and a first taste of functions. Capstone: Treasure Hunt text adventure.

Big question: How do loops, chance and lists turn a program into a game?

Assessment

Coding challenges are checked on the server by running the learner's program. Quick checks give instant feedback. Written answers and projects are reviewed by the teacher against 4-level rubrics (Beginning · Developing · Secure · Excellent).

What you need

- Laptop or Chromebook with a modern browser and a keyboard
- The built-in Kitabash Python Lab (runs in the browser, no install or account needed)
- Paper and pencil for planning maps and quizzes

Before you start

- None. Learners should be able to type on a keyboard and read simple English instructions.

Safety and privacy

Programs never ask for real personal information: use first names or made-up names only, and never type phone numbers, addresses, school names or passwords into a program or the chat. Breakout rooms are supervised by the teacher. Learners under 18 use the AI tutor only with guardian consent.

What comes next

After this course: **Python Game Coders · Level 2 (functions, dictionaries, turtle graphics, bigger games)**.

Kitabash original curriculum.